

FAKE NEWS DETECTOR

A MINI-PROJECT REPORT

submitted by

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SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY

(Affiliated to APJ Abdul Kalam Technological University)

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CERTIFICATE

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DECLARATION

We undersigned hereby declare that the project report on 'Fake News Detector', submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Mr Safeer P S, Associate Professor, Department of Computer Science, SSET. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not mis-represented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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ABSTRACT

Fake news has become a serious problem in the digital era, spreading rapidly through social media and online platforms. It often misleads people, influences public opinion, and creates social and political issues. Manual verification of news is time-consuming and cannot handle the large volume of information available online. Therefore, an automated system is required to detect fake news efficiently and accurately. This project proposes a Fake News Detector using Machine Learning and Natural Language Processing techniques. The system analyzes news articles and classifies them as Real or Fake based on textual content. A dataset containing labeled news articles is used for training and testing the model. The text data is preprocessed using tokenization, stopword removal, and stemming. Feature extraction is performed using TF-IDF vectorization to convert text into numerical features. Two machine learning models, Naive Bayes and Passive Aggressive Classifier, are used for classification. These models are trained to identify patterns and improve prediction accuracy. A Flask-based web application is developed to provide a simple user interface. Users can input news text and receive instant predictions along with a confidence score. The system ensures fast and reliable detection of fake news in real time. This project demonstrates the effectiveness of machine learning in combating misinformation.

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CHAPTER 1

INTRODUCTION

Fake news is a growing concern in today's digital world, with misinformation spreading rapidly through social media, news platforms, and online communication channels. Many people are exposed to false or misleading information on a daily basis, often without the ability to verify its authenticity. The absence of real-time verification systems and the ease of sharing unverified content worsen these issues. This highlights the urgent need for technology-driven solutions to detect and prevent the spread of fake news. With advancements in Machine Learning and Natural Language Processing (NLP), digital solutions now play a key role in identifying and filtering fake news. Automated systems can analyze large volumes of textual data and detect patterns that differentiate real news from fake content. The Fake News Detector is designed to leverage these technologies, ensuring accurate and efficient news verification. It provides a platform where users can input news content and instantly receive predictions about its authenticity. The system includes a classification mechanism that processes news articles using preprocessing techniques such as tokenization, stopwords removal, and stemming. It uses TF-IDF vectorization for feature extraction and applies machine learning algorithms like Naive Bayes and Passive Aggressive Classifier to classify news as real or fake. The system also provides a confidence score for each prediction, helping users understand the reliability of the result. This report analyzes the Fake News Detector system, its development, functionalities, and impact. It also examines the challenges of misinformation and how technology helps address them through reliable detection methods. By implementing machine learning-based solutions, the system helps reduce the spread of fake news and promotes the use of trustworthy information in the digital space.

CHAPTER 2

LITERATURE REVIEW

Ensuring the detection of fake news using technological advancements has become a key focus in recent years. Several studies and research papers have explored the use of Machine Learning, Natural Language Processing (NLP), and Deep Learning techniques to identify and classify fake news. This literature review summarizes existing research on fake news detection systems.

Paper [1]: "Fake News Detection on Social Media using Machine Learning" (Shu, Sliva, Wang, Tang, & Liu, 2017)

This study presents a machine learning-based approach to detect fake news on social media platforms. The system uses content-based and social context-based features for classification. The authors highlight the importance of combining multiple features to improve detection accuracy and reduce misinformation spread.

Paper [2]: "Detecting Fake News in Social Media Networks" (Castillo, Mendoza, & Poblete, 2011)

This paper focuses on analyzing user behavior and content credibility in social media. It uses classification algorithms to identify trustworthy and untrustworthy information. The authors emphasize that both content and user interaction patterns are essential for effective fake news detection.

Paper [3]: "CSI: A Hybrid Deep Model for Fake News Detection" (Ruchansky, Seo, & Liu, 2017)

This paper introduces a hybrid model that combines text analysis, user response, and source reliability. The model improves detection performance by integrating multiple aspects of news data. Highlight the effectiveness of hybrid approaches in handling complex fake news patterns.

Paper [4]: "Deception Detection and Fake News Identification" (Rubin, Chen, & Conroy, 2015)

This research focuses on linguistic and psychological analysis of fake news. It examines writing style, emotional tone, and language patterns to detect deception. The authors emphasize that linguistic cues play a major role in identifying misleading information.

Paper [5]: "Fake News Detection using Machine Learning Techniques" (Ahmed, Traore, & Saad, 2017)

This study applies various machine learning algorithms such as Naive Bayes and Support Vector Machines for classification. The authors compare different models and conclude that machine learning provides efficient and scalable solutions for fake news detection.

Paper [6]: "Fake News Detection using Deep Learning" (Wang, 2017) This paper explores deep learning techniques for fake news classification. Neural networks are used to capture complex patterns in textual data. The study highlights that deep learning models improve accuracy but require large datasets and computational resources.

Paper [7]: "Natural Language Processing for Fake News Detection" (Kaliyar, Goswami, & Narang, 2021)

This research focuses on NLP-based feature extraction methods such as TF-IDF and word embeddings. The authors emphasize that proper preprocessing and feature selection significantly improve model performance and classification accuracy.

The reviewed literature highlights the growing importance of machine learning and NLP-based approaches in detecting fake news. Key advancements include hybrid models, deep learning techniques, and feature extraction methods like TF-IDF. Future research should focus on improving model accuracy, handling multilingual data, and integrating real-time detection systems to enhance the effectiveness of fake news detection solutions.

2.1 Inference

The Fake News Detector system provides a simple and user-friendly interface designed to ensure easy interaction for users. The interface is developed as a web-based application using Flask, allowing users to access the system through a browser without the need for installation. The design focuses on simplicity, clarity, and quick response to enhance user experience.

The main interface consists of a text input area where users can enter or paste the news article or content they want to verify. A submission button is provided to process the input and send it to the backend system for analysis. Once the user submits the text, the system processes the input using machine learning models and displays the result on the same page.

The output section of the interface clearly shows whether the news is Real or Fake, along with a confidence score indicating the reliability of the prediction. The result is displayed in a clean and readable format, ensuring that users can easily understand the outcome. The interface is designed to provide quick feedback, enabling real-time verification of news content.

Additionally, the interface maintains a structured layout with proper headings, buttons, and spacing to improve usability. It ensures smooth navigation and responsiveness across different devices. The simplicity of the interface makes it accessible to all users, even those without technical knowledge.

Overall, the interface of the Fake News Detector is designed to provide an efficient, interactive, and user-friendly platform for detecting fake news, making the system practical for everyday use.

CHAPTER 3

PROBLEM STATEMENT

In today's digital world, the spread of fake news has become a serious and growing concern. With the rapid rise of social media platforms and online news sources, misinformation can spread quickly and influence public opinion, political decisions, and social harmony. Many users are exposed to false or misleading information daily, often without the means to verify its authenticity. Traditional methods of fact-checking, such as manual verification and expert analysis, are time-consuming and unable to handle the vast amount of information generated every second. As a result, fake news continues to spread unchecked, leading to confusion, misinformation, and a lack of trust in reliable sources.

The Fake News Detector addresses this issue by providing an automated, technology-driven solution for identifying false information. By leveraging Machine Learning and Natural Language Processing (NLP), the system analyzes news content and classifies it as real or fake in real time. The system enables users to input news text and receive instant predictions along with a confidence score, helping them make informed decisions. Additionally, the system ensures fast processing, scalability, and ease of use through a web-based interface. By reducing the spread of misinformation and promoting reliable information verification, the Fake News Detector contributes to creating a more informed and trustworthy digital environment.

3.1 Objective

The objective is to develop an efficient machine learning-based system that automatically detects and classifies news articles as real or fake while enabling quick and reliable verification for users.

Features include machine learning-based detection to classify news based on learned patterns, real-time news verification for instant analysis and prediction, NLP-based text processing such as tokenization, stop-word removal, and stemming for effective data handling, feature extraction using TF-IDF to convert text into numerical form and improve accuracy, a user-friendly web interface for easy input and clear results with confidence scores, and reduction of misinformation spread by providing a reliable tool to verify information before sharing.

CHAPTER 4

COMPARISON

Feature	Alt News	Check-It	XFake System	Fake News Alert	NewsGuard / Our.News	Fake News Detector 101	Antidote Tool
Year	2017	2019	2019	2017	2018	2026	2018
Developer	Pratik Sinha, Mohammed Zubair	Paschalides et al.	Fan Yang et al.	Brian Feldman	Media & research organizations	AI-based platform	DSPolitical
Real-time Detection	✗	✓	⚠	✗	⚠	✓	✗
Manual Fact-checking	✓	✗	✗	✗	✗	✗	✗
AI-based Detection	✗	✓	✓	✗	⚠	✓	✗
Explainable Results	✗	⚠	✓	✗	⚠	✓	✗
Browser Integration	✗	✓	✗	✓	✗	✗	✗
Database of Fake Sources	✗	⚠	✗	✓	✓	⚠	✓
Multilingual Support	✓	⚠	✗	✗	⚠	⚠	✗
Limitations	⚠	⚠	⚠	⚠	⚠	⚠	⚠

CHAPTER 5

SYSTEM DESCRIPTION

The Fake News Detection System is a web-based application developed to identify whether a given news article is real or fake using machine learning and natural language processing techniques. The system provides a simple and efficient interface where users can input news content and receive an immediate classification result along with a confidence score.

The system follows a client-server architecture. The frontend is developed using HTML and CSS, providing a user-friendly interface for entering news text and viewing results. The backend is implemented using the Flask framework in Python, which handles user requests, processes input data, and communicates with the machine learning models. The backend acts as the core component where all computations and predictions take place.

When a user submits news text, the system performs preprocessing steps such as cleaning and formatting the input. The processed text is then transformed into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) vectorization technique. This method converts textual data into feature vectors by assigning importance to words based on their frequency and relevance within the dataset.

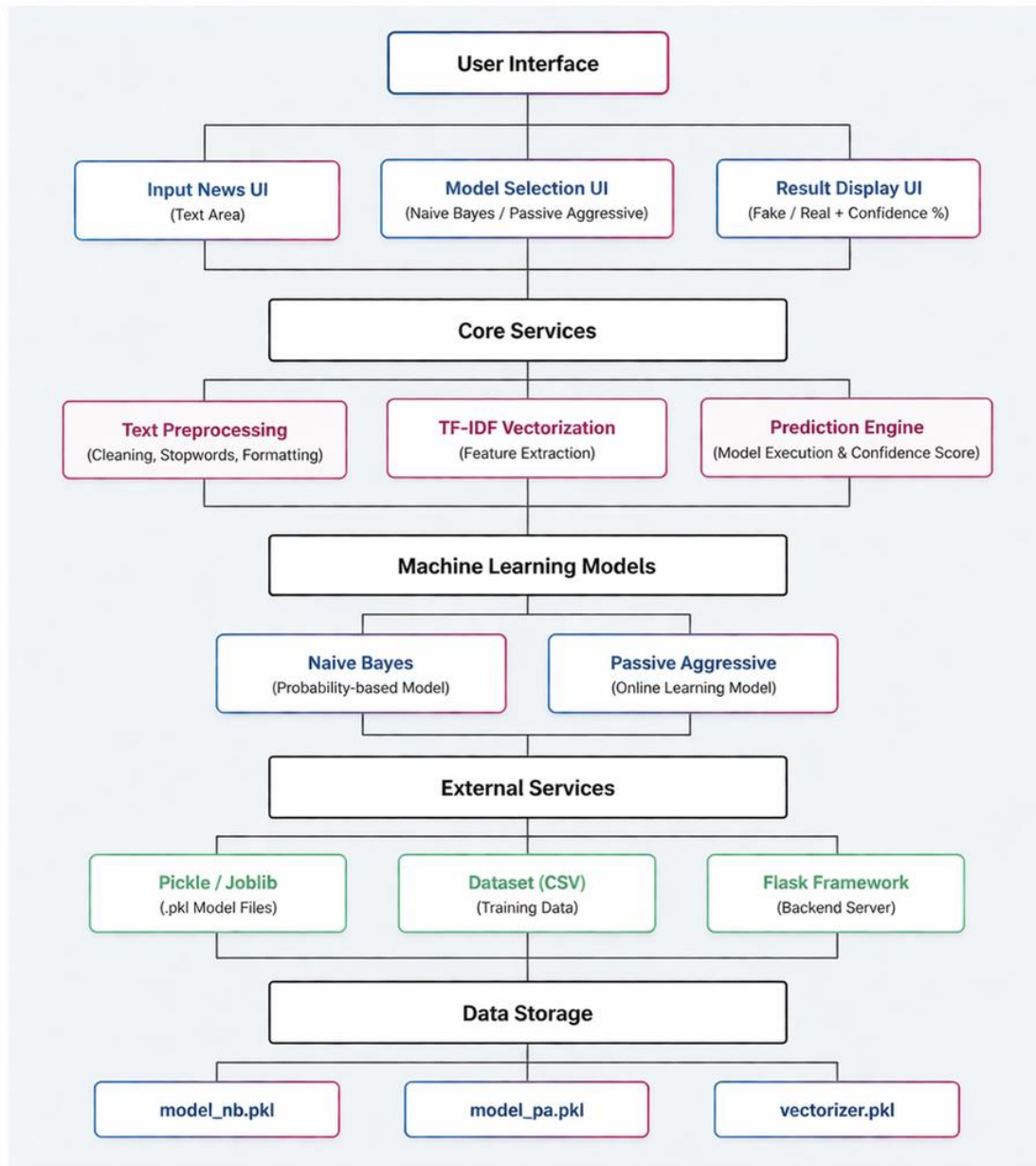
The system uses two machine learning models for classification: Naive Bayes and Passive Aggressive. The Naive Bayes classifier is a probabilistic model that calculates the likelihood of the news belonging to a particular class (real or fake) based on word distributions. The Passive Aggressive classifier is an online learning algorithm that updates its model only when it makes an incorrect prediction, making it efficient for handling large and continuously evolving datasets.

After processing the input through the selected model, the system generates a prediction indicating whether the news is real or fake. In addition, a confidence score is provided to indicate the reliability of the prediction. For the Naive Bayes model, the confidence score is calculated using probability estimates. For the Passive Aggressive model, an approximate confidence value is used since it does not inherently provide probability outputs.

The trained models and the TF-IDF vectorizer are stored as serialized files in .pkl format. These files are loaded at runtime, allowing the system to perform predictions quickly without retraining the models each time.

Overall, the Fake News Detection System provides a practical and efficient solution for identifying misinformation. By integrating machine learning techniques with a web-based interface, the system ensures fast, accurate, and user-friendly detection of fake news content.

5.1. SYSTEM ARCHITECTURE



CHAPTER 6

SYSTEM DESIGN

6.1. Software

6.1.1. Tools Used

The Fake News Detection System is developed using a combination of frontend, backend, and machine learning technologies. Each tool plays a specific role in ensuring the system is efficient, accurate, and user-friendly.

1. Frontend Development

HTML and CSS

- Used for designing the structure and layout of the web application
- HTML creates elements like text area, buttons, and result sections
- CSS is used for styling such as colors, fonts, spacing, and layout

JavaScript

- Adds interactivity to the web page
- Handles user actions like button clicks and form submissions
- Helps update content dynamically without reloading the page

CSS Animations

- Provides smooth transitions and visual effects
- Enhances user experience through hover effects and animations

Responsive Design

- Ensures the application works on different devices
- Supports mobile, tablet, and desktop screens

2. State Management

JavaScript DOM Manipulation

- Manages user input and updates the UI dynamically
- Allows displaying results without refreshing the page

Form Handling

- Captures user input from the text box
- Sends data to the backend using HTTP requests

3. Backend and API Development

Flask (Python)

- Acts as the backend framework
- Handles user requests and sends responses
- Connects frontend with machine learning models

Jinja2 Templates

- Used for dynamic HTML rendering
- Displays prediction results and confidence scores

RESTful Routing

- Manages GET and POST requests
- Ensures smooth communication between frontend and backend

4. Machine Learning and Processing

TF-IDF Vectorizer

- Converts text data into numerical form
- Assigns importance to words based on frequency
- Helps models understand textual data

Naive Bayes Classifier

- Probability-based classification algorithm
- Calculates likelihood of news being real or fake
- Provides confidence score using probabilities

Passive Aggressive Classifier

- Online learning algorithm
- Updates model only when prediction is incorrect
- Efficient for large datasets

Text Preprocessing

- Cleans input data before processing
- Removes punctuation and stop words
- Converts text to lowercase for consistency

5. Model Storage and Serialization

Pickle (.pkl files)

- Stores trained machine learning models
- Saves model structure and learned parameters

Joblib

- Loads models efficiently
- Faster for large datasets compared to pickle

Pre-trained Models

- Models are trained beforehand
- Used directly during prediction
- Reduces processing time

6. Database and Data Handling

CSV Dataset

- Contains real and fake news data
- Used for training machine learning models

File-based Storage

- Stores dataset and model files locally
- No separate database required

7. Prediction and Output System

Model Selection

- User selects between Naive Bayes and Passive Aggressive
- Selected model is used for prediction

Prediction Engine

- Processes vectorized input
- Generates classification result

Confidence Score Calculation

- Uses probability values for Naive Bayes
- Shows reliability of prediction as percentage

8. User Interface Features

Text Input Area

- Allows user to enter news content

Model Selection Dropdown

- Lets user choose classification model

Result Display Panel

- Shows prediction (Real or Fake)

Confidence Indicator

- Displays confidence percentage

9. UI/UX Enhancements

CSS Animations

- Smooth scrolling and transitions

Interactive Elements

- Buttons, hover effects, and dynamic updates

10. Development Tools

Python

- Main programming language

VS Code

- Development environment

Git and GitHub

- Version control and collaboration

Virtual Environment (venv)

- Manages project dependencies

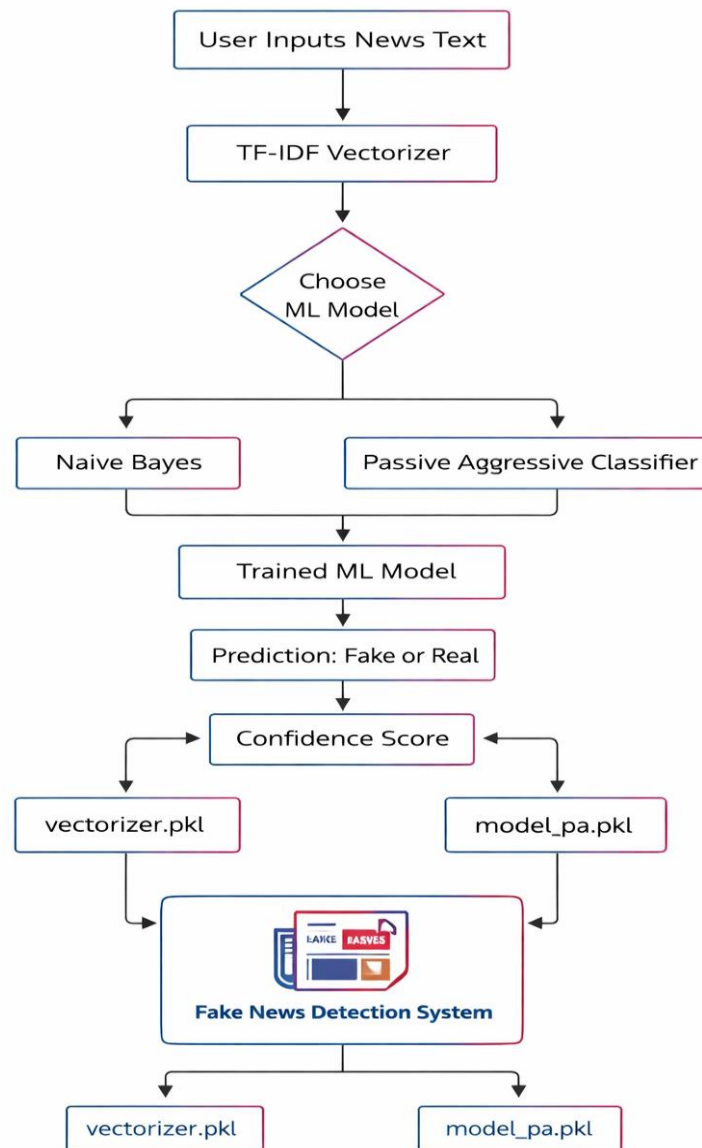
11. System Workflow Summary

- User enters news text
- Data is sent to Flask backend
- Text is preprocessed and vectorized
- Selected model analyzes the input
- Prediction (Fake or Real) is generated
- Confidence score is displayed

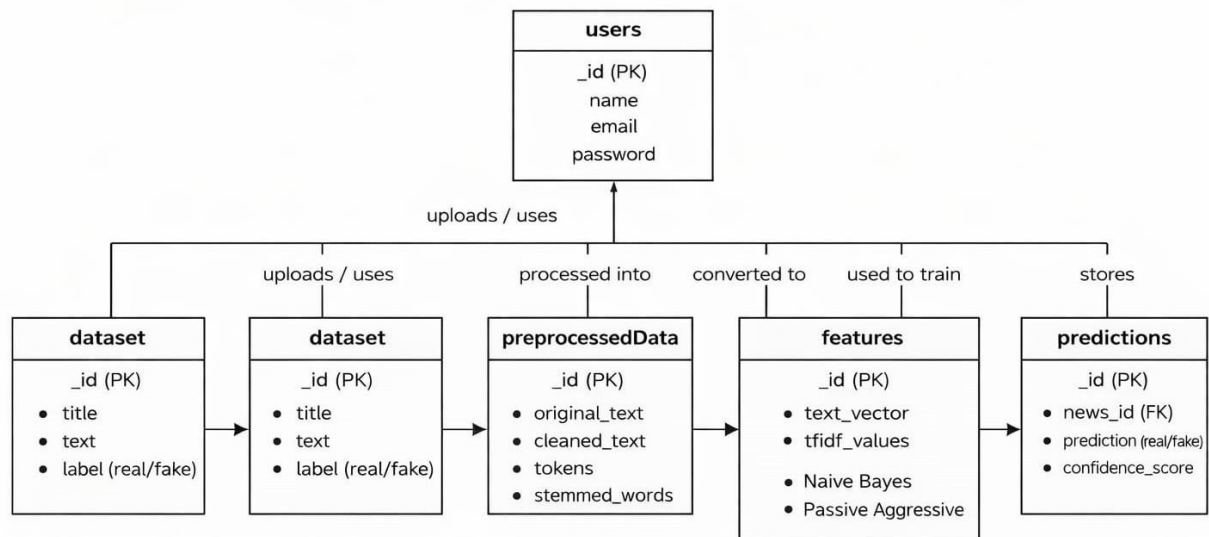
Conclusion

The system integrates frontend technologies with backend processing and machine learning models to provide accurate and fast fake news detection. It ensures a user-friendly experience while maintaining high efficiency and reliability.

6.1.2. Flow Chart



6.1.3. Dataset



Database Collections

6.1.4. Working of the project

The Fake News Detection System is a web-based application that uses machine learning techniques to classify news content as Real or Fake. It integrates Natural Language Processing (NLP), TF-IDF vectorization, and classification algorithms to provide accurate predictions along with confidence scores.

Below is a detailed explanation of how each component works.

1. App Interface & Core Framework

How it Works:

Users interact with the system through a simple web interface.

The interface allows users to input news text and select a machine learning model.

The frontend sends the input data to the backend for processing and displays the result.

Tools Used:

Frontend: HTML, CSS

Framework: Flask (Python)

2. Input Processing Module

How it Works:

The user enters a news article or headline in the input text box.

The input is captured through a form and sent to the Flask backend using a POST request.

The text is prepared for further processing.

Tools Used:

Flask (request handling)

HTML forms

3. Text Preprocessing & Feature Extraction

How it Works:

The input text is transformed into numerical format using TF-IDF Vectorization.

TF-IDF assigns importance to words based on frequency and uniqueness.

This conversion allows machine learning models to understand textual data.

Tools Used:

TF-IDF Vectorizer (Scikit-learn)

Pre-trained vectorizer.pkl file

4. Model Selection Module

How it Works:

The system allows users to choose between two machine learning models:

Naive Bayes Classifier

Passive Aggressive Classifier

Based on user selection, the corresponding model is loaded and used for prediction.

Tools Used:

Scikit-learn models

Stored .pkl model files

5. Prediction Engine

How it Works:

The selected model receives the vectorized input.

The model analyzes patterns learned during training.

It predicts whether the news is Fake or Real.

Tools Used:

Naive Bayes (model_nb.pkl)

Passive Aggressive (model_pa.pkl)

6. Confidence Score Calculation

How it Works:

If the model supports probability prediction:

`predict_proba()` is used to calculate confidence.

The highest probability is converted into percentage.

If not supported (like Passive Aggressive):

A default confidence score is assigned or derived from decision function.

Formula: $\text{Confidence} = \max(\text{probability values}) \times 100$

Tools Used:

Scikit-learn probability functions

7. Output Display Module

How it Works:

The result (Fake or Real) is displayed on the webpage.

Confidence percentage is shown along with the prediction.

The UI dynamically updates without clearing user input.

Tools Used:

Flask (`render_template`)

HTML + CSS for styling

8. Model Storage & Loading

How it Works:

Pre-trained models and vectorizer are stored as .pkl files.

These files are loaded when the application starts.

This avoids retraining and ensures faster performance.

Files Used:

`model_nb.pkl`

`model_pa.pkl`

`vectorizer.pkl`

Tools Used:

`pickle` / `joblib`

9. Overall System Workflow

User enters news text

Text is sent to backend

TF-IDF converts text to numeric form

User-selected model processes input

Model predicts Fake or Real

Confidence score is calculated

Result is displayed to user

Conclusion

The Fake News Detection System effectively combines machine learning and web technologies to identify misleading information. By using TF-IDF for feature extraction and classifiers like Naive Bayes and Passive Aggressive, the system provides fast and reliable predictions. The integration of a user-friendly interface with backend processing ensures efficient detection and enhances user awareness about misinformation.

CHAPTER 7

RESULTS

THE DAILY DETECTOR · EST. 2026

FAKE NEWS DETECTOR

Enter news here...

Naive Bayes

ANALYZE NEWS

BBC TIMES OF INDIA CNN INTERNATIONAL ZEE MEDIA ASIANET NEWS MANORAMA 24 X 7 UPDATES

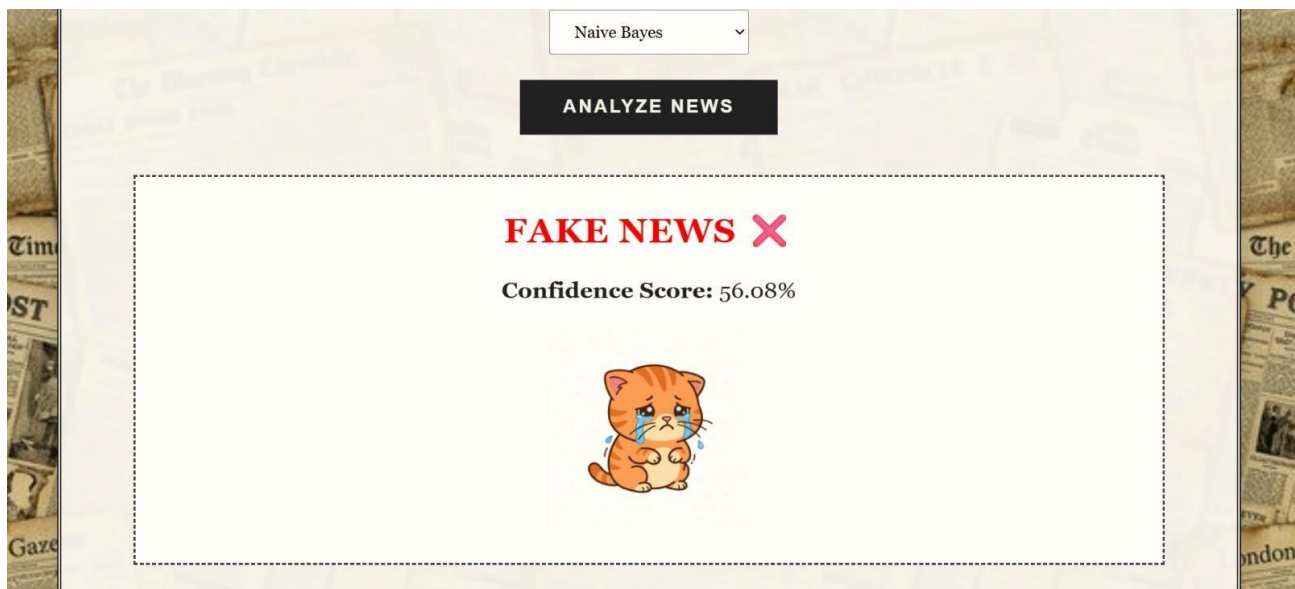
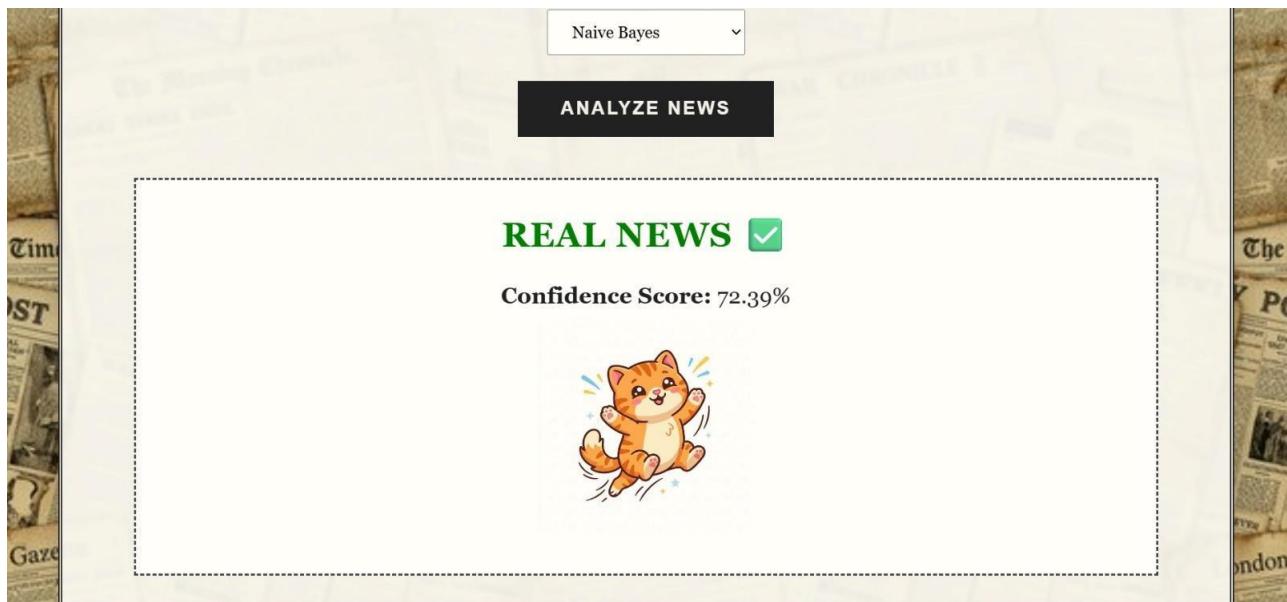
TOP STORIES

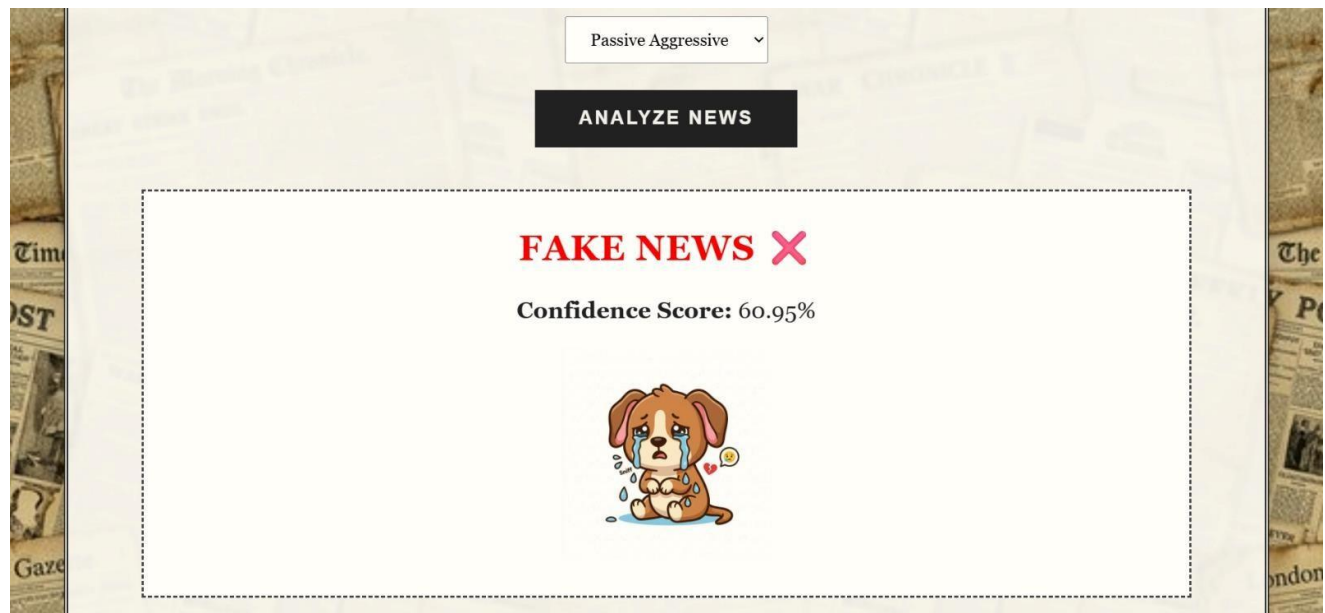


Times of India: Global Markets Rally
Comprehensive coverage of today's financial trends and economic impacts worldwide.



BBC: Breakthroughs in Renewable Energy
New technologies are paving the way for a more sustainable future.





CHAPTER 8

CONCLUSION & FUTURE SCOPE

8.1. Conclusion

In conclusion, the Fake News Detector is a robust and efficient web-based application developed to address the growing issue of misinformation in the digital age. With the rapid spread of news through online platforms and social media, the need for reliable verification systems has become increasingly important. This system leverages the power of Machine Learning and Natural Language Processing (NLP) techniques to analyze and classify news content as real or fake with a high degree of accuracy. By integrating preprocessing techniques, TF-IDF feature extraction, and classification algorithms such as Naive Bayes and Passive Aggressive Classifier, the system ensures fast and dependable results.

The architecture of the system is designed to be simple yet effective, combining a user-friendly frontend interface with a powerful backend processing unit. Users can easily input news content through the interface, select a preferred model, and receive instant predictions along with a confidence score. This real-time response enhances usability and makes the application suitable for everyday use by individuals with varying levels of technical expertise. The use of pre-trained models stored in serialized formats further improves efficiency by eliminating the need for repeated training during runtime.

Beyond its technical capabilities, the Fake News Detector serves a significant social purpose by helping to reduce the spread of false information. In an era where misinformation can influence public opinion and decision-making, this system acts as a valuable tool for promoting awareness and encouraging critical evaluation of online content. By providing quick and accessible verification, it empowers users to make informed decisions and discourages the sharing of unverified or misleading news.

Furthermore, the system demonstrates how modern technologies can be effectively utilized to solve real-world problems. Its lightweight design, fast processing, and integration of machine learning techniques make it both practical and scalable. The modular structure of the system also allows for future enhancements, such as incorporating advanced deep learning models, multilingual support, and real-time data integration.

Overall, the Fake News Detector represents a meaningful step toward building a more trustworthy digital ecosystem. By combining technological innovation with user-centric design, the system not only delivers accurate predictions but also contributes to maintaining information integrity. It stands as an effective solution in combating fake news and highlights the importance of leveraging technology to address critical societal challenges.

8.2. Future Scope

The proposed Fake News Detection System can be further enhanced by incorporating advanced deep learning techniques and expanding its functional capabilities to improve accuracy, scalability, and real-world applicability. One of the major future enhancements includes the integration of deep learning models such as Long Short-Term Memory (LSTM) networks and Bidirectional Encoder Representations from Transformers (BERT). These models are capable of understanding contextual relationships between words in a much deeper manner compared to traditional machine learning algorithms. By leveraging such models, the system can significantly improve its ability to detect subtle patterns, sarcasm, and complex linguistic structures often present in fake news content, thereby increasing overall prediction accuracy.

Another important enhancement involves extending the system to support multilingual news analysis. Currently, the system primarily processes news content in a single language. However, fake news is a global issue that spreads across multiple languages and regions. By integrating multilingual Natural Language Processing techniques and pre-trained language models, the system can be adapted to analyze and classify news articles in various languages. This will make the platform more inclusive and accessible to a broader audience, allowing users from different linguistic backgrounds to verify information effectively.

The system can also be improved by incorporating real-time data integration through news APIs and social media platforms. Fake news often spreads rapidly through platforms such as social media, making real-time detection crucial. By connecting the system with live news feeds and social media streams, it can continuously monitor and analyze trending content. This will enable early detection of misinformation and provide users with timely alerts, thereby helping to reduce the spread of false information.

In addition, the development of browser extensions and mobile applications can greatly enhance user accessibility and convenience. A browser extension would allow users to instantly verify the authenticity of online articles while browsing, whereas a mobile application would provide on-the-go access to the detection system. These enhancements will ensure that the system is easily available across different platforms, increasing user engagement and usability.

Furthermore, incorporating a user feedback mechanism will play a vital role in improving the system's performance over time. Users can provide feedback on predictions, which can be used to retrain and refine the models. This continuous learning approach ensures that the system adapts to new patterns of misinformation and remains up-to-date with evolving fake news strategies.

Finally, implementing continuous model training and updating mechanisms will ensure that the system maintains high accuracy and reliability. As new datasets become available, the models can be retrained periodically to improve their performance. This will help the system remain robust and effective in detecting emerging trends in fake news.

Overall, these future enhancements aim to transform the Fake News Detection System into a more intelligent, scalable, and user-friendly platform capable of addressing the growing challenges of misinformation in the digital age.

CHAPTER 9

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